Monte Carlo - a completely probabilistic method

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1 Introduction

The Monte Carlo method was developed by a team of scientists working on the Manhattan Project. It involves continuously drawing samples and checking whether they meet a specific criterion. It operates on the principle of the Law of Large Numbers, which states that the more drawings are performed, the closer the average result will become to the expected value.

2 Problem description

In this case, I decided to solve the problem of calculating the volume of a cone inscribed in a cube using random sampling. The radius of the cone is equal to half the cube's side, i.e., 4, and the height of the cone is 8. When selecting random points for a cone, the problem of decreasing height with increasing radius arises, but these two variables can be interdependent using similar triangles. After drawing the height through the cross-section of the cone, we can find two similar triangles with one base being the radius and the other being the radius at a given height. We then determine the relationship where ABC is the large triangle and DEC is the small triangle.

$$\frac{AB}{DE} = \frac{AC}{DC}$$

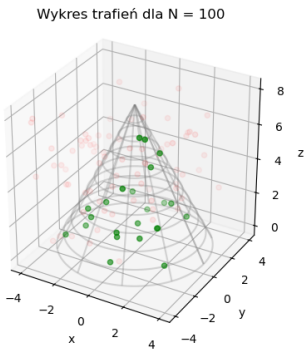
Which we then transform into a dependency:

$$\frac{R(z)}{r} = \frac{H - z}{H}$$

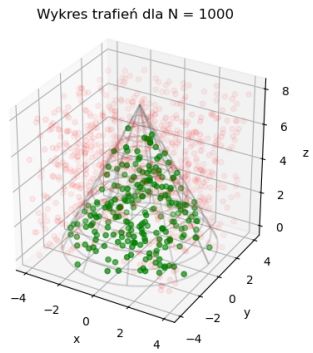
$$R(z) = r * (1 - \frac{z}{H})$$

Where R is our radius function versus height.

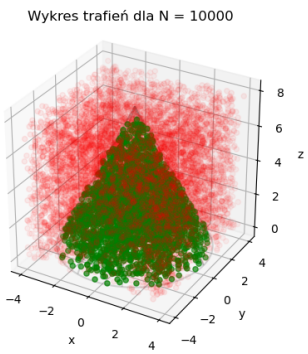
3 Point distribution



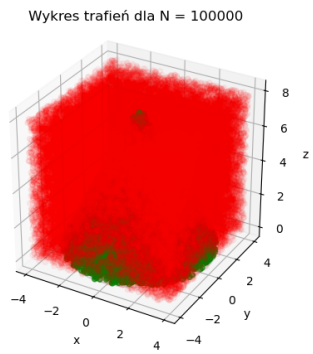
(a) N = 100



(b) N = 1000



(c) N = 10000

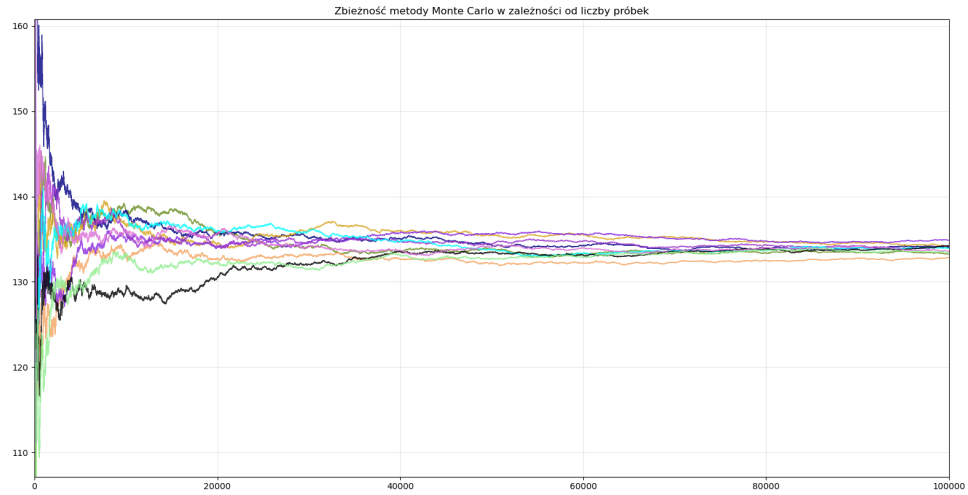


(d) N = 100000

Rysunek 1: Wykresy losowań w zależności od ilości próbek.

In the case of these four plots, it's clear that the values begin to fill the cone relatively evenly for N greater than 10,000 samples. For sampling values smaller than this, there's a clear predominance of red points that missed our cone. This will result in a large deviation from the expected value for a small number of samples. This suggests that Monte Carlo should find a "golden mean"because, with unlimited computing power, we could theoretically approximate our value to an unlimited number of decimal places.

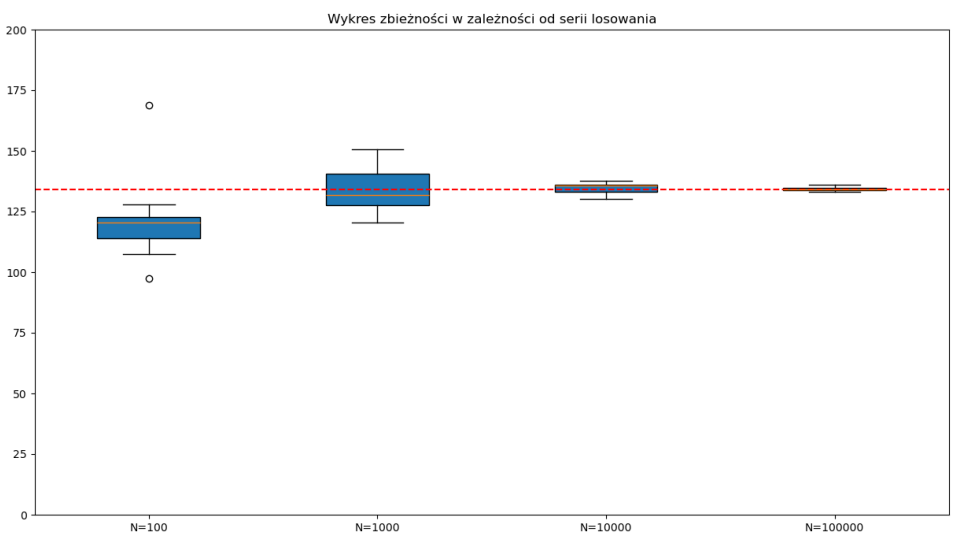
4 Convergence chart



Rysunek 2: Convergence chart MC

By analyzing the convergence graph, it can be clearly seen that the law of large numbers is at work here: the more N tends to infinity, the closer to the theoretical value our results are.

5 Boxplot



Rysunek 3: Boxplot chart depending on the number of samples

As the boxplot shows, with increasing sample size, the interquartile range decreases and outliers disappear. It appears that for sample sizes higher than 10^6, the time complexity becomes unprofitable in terms of decimal precision